

Ultrasonic Sensor Controlled Bulb Using Relay

This project explains how an ultrasonic sensor can automatically control a bulb using a relay module and a microcontroller such as Arduino UNO. The ultrasonic sensor detects the distance of an object or person. When someone comes near the sensor, the relay turns ON the bulb. When the person moves away, the relay turns OFF the bulb.

Required Components

Component	Purpose
Arduino UNO	Controls the system
Ultrasonic Sensor (HC-SR04)	Detects distance
Relay Module	Switches the bulb ON/OFF
Bulb	Output load
Jumper Wires	Connections
Power Supply	Provides power

Working Principle

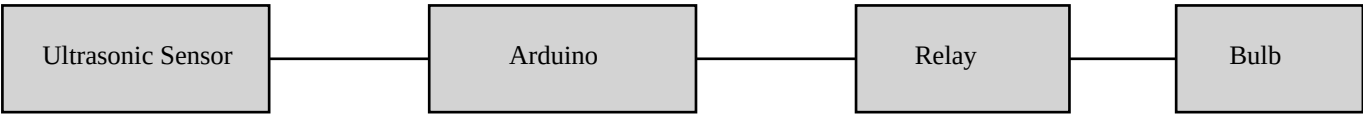
The HC-SR04 ultrasonic sensor sends ultrasonic waves and receives the reflected signal. The Arduino calculates the distance using the echo time.

- If the detected distance is less than a fixed value (for example 20 cm), the Arduino activates the relay.
- The relay closes the circuit and the bulb turns ON.
- If the distance is greater than the set value, the relay turns OFF and the bulb switches OFF.

Connections

Ultrasonic Sensor Pin	Arduino Pin
VCC	5V
GND	GND
TRIG	Digital Pin 9
ECHO	Digital Pin 10
Relay IN	Digital Pin 7

Block Diagram



Arduino Program

```
#define trigPin 9
#define echoPin 10
#define relayPin 7

long duration;
int distance;

void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(relayPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {

  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);

  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  duration = pulseIn(echoPin, HIGH);

  distance = duration * 0.034 / 2;

  Serial.println(distance);

  if (distance < 20) {
    digitalWrite(relayPin, HIGH);
  } else {
    digitalWrite(relayPin, LOW);
  }

  delay(500);
}
```

Applications

- Automatic room lighting
- Smart home systems
- Energy saving applications
- Contactless switching systems

Advantages

- Saves electricity
- Automatic operation
- Easy to build and low cost
- Useful for smart automation